

Computer Networks

CS - (Fall 25)

Assignment-2

Marks 30

Submission *deadline* **26th October 2025 (11:59 PM)**

Instructions:

- *This is an individual Assignment. Demos will start soon after submission date, **missing demo will result in straight zero marks.***
 - ***All parties involved in any kind of plagiarism/cheating will be given zero marks in all the assignments.***
 - ***Assignment deadline will not be extended***
 - ***Late submissions will be discarded so submit your assignment on-time***
 - *You must follow below submission guidelines, otherwise your submission won't be accepted*
 - *create a new directory*
 - *change its name to the following format*
YOURSECTION_ROLLNUMBER_NAME.
E.g. A_20I-0012_Muhammad Ahamd
 - *put all your files into this newly created directory (scan pdf for theoretical questions)*
 - *Compress the directory into a compressed .zip file*
 - *Submit it on google classroom*
 - *Understanding the problem is also part of the assignment.*
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Q1. RDT Design: Eliminating Ambiguity with Corruption and Loss (10 Marks)

Design an rdt variant over a channel that (a) may corrupt data/ACKs, (b) may lose data/ACKs, and (c) may arbitrarily delay but not reorder. Your design must be NAK-free and stop-and-wait. Specify:

1. Sender FSM states, events, actions.
2. Receiver FSM states, events, actions.
3. Exact ACK contents (what sequence number is ACKed and why).
4. How sender handles duplicate ACKs and timeouts to avoid delivering duplicates to the app.
5. Prove that at-most-once delivery holds given these assumptions.

Q2. Stop-and-Wait Utilization and Scaling (4 Marks)

A 2,500 km fiber link (speed of light in fiber $\approx 2 \times 10^8$ m/s) connects a sender and receiver. Link rate $R = 10$ Gbps. $MSS = 12,000$ bits of payload per packet. Stop-and-wait is used. Compute:

1. One-way propagation delay, RTT.
2. Transmission time L/R .
3. Sender utilization $U = (L/R) / (RTT + L/R)$.
4. The factor increase in utilization if a 64-packet pipeline is allowed (assume perfect channel and immediate ACKs upon each packet, cumulative acceptable).
Round to three significant digits.

Q3. TCP Seq/Ack Accounting Under Reordering at the Receiver (6 Marks)

A TCP sender transmits three segments with payload sizes 1000 bytes each, initial sequence number $ISN_A = 10,000$. The network delivers them to the receiver in order of second, third, then first segment. The receiver buffers out-of-order data. For each arrival, state:

1. Receiver's cumulative ACK number sent back.
2. The highest contiguous byte received at that moment.
Assume receiver acknowledges cumulatively the next expected byte only (no SACK), and buffers out-of-order data.

Q4. GBN vs SR: Transmission Counts and Completion Time (Specified Loss/Delay) (5 Marks)

You have the following situation:

- One-way delay d ; negligible serialization; window $W=4$; timeout $T=4d$; need to deliver packets P_0 to P_5 .
- Data losses: P_1 and P_3 lost; others arrive.
- ACK delays: ACK_0 and ACK_2 delayed by $3d$; others delayed by d .

For each protocol, compute total data transmissions and the time when final in-order delivery completes.

Q5. SR Out-of-Order Trace Problem (5 Marks)

A sender sends packets 0–4 to the receiver, however packets 1 and 3 are lost. Show receiver's buffer status and ACK pattern at each step until all delivered. Fill in the following table:

Step	Packet Arrives	Receiver State (Buffer)	Sent ACKs	Delivered
1				
2				
3				
4				
5				